Invasive spinal cord computer interfaces for medicine and human capability

A mechanistic review and hypothesis paper on motor learning and sustained physiological control

Research and hypothesis paper | Evidence through 30 September 2026

Abstract

Invasive spinal cord–computer interfaces could become a platform for restoring movement, supplying bodily sensory feedback, regulating selected autonomic functions, and accelerating the acquisition of some motor skills. The strongest argument for these futures is a constructive causal chain: spinal activity contains information about bodily state; implanted stimulation can alter relevant circuits; feedback can coordinate stimulation with ongoing behavior; and activity-dependent plasticity can preserve some changes after assistance ends. Several links already have experimental support, although the strength and scope of that support vary substantially.

This paper examines what could be possible under known physics, mathematics, and biology while setting aside manufacturing, commercialization, surgical practicality, and ethical barriers. It develops two principal hypotheses. First, spinal recordings from experts could inform a learner-specific, closed-loop training policy that improves retained motor learning. Second, an implanted controller could maintain, repeat, or change selected involuntary physiological states for longer than an unaided individual can voluntarily manage. Both proposals become scientifically meaningful when the target function, accessible neural pathways, feedback variables, and success criteria are specified.

The analysis distinguishes a known contradiction with biology from an untested implementation. It also distinguishes a plausible mechanism from evidence for a particular effect size. A 1.5-fold or twofold reduction in training time is a legitimate experimental target, but cannot be calculated from present spinal-interface studies alone. Similarly, evidence that stimulation evokes an organ response supports access to that response; selective stabilization or postponement requires additional control authority.

The resulting forecast is favorable for targeted bodily interfaces, including adaptive rehabilitation, sensory prostheses, physiological regulation, and embodied training. A universal spinal implant for all cognition, all organs, or native full-sensory virtual reality requires access and causal mechanisms beyond those established for spinal pathways. These are anatomical and information constraints, not arguments from the historical absence of a product.

Article type: narrative evidence review and hypothesis paper. Evidence cutoff: 30 September 2026. Literature checked: 1 October 2026. No original human or animal experiment is reported. Numerical learning curves are explicitly illustrative.

Keywords: spinal cord–computer interface; invasive neural interface; motor learning; autonomic control; neural plasticity; sensory feedback; brain–spine interface.

Contents

Abstract 1
1 Scope and method 3
2 What theoretical possibility means 4
3 The spinal cord as a processing and control system 5
4 Interface architectures and causal access 6
5 Evidence that the spinal cord can be read 7
6 Evidence that spinal stimulation changes useful outputs 8
7 Physics and information constraints 9
8 A mathematical model of task control 10
9 Hypothesis one Expert informed skill acceleration 11
10 Biological support for the learning hypothesis 12
11 What a learning multiplier would mean 13
12 Candidate learning applications 14
13 How to test expert informed learning 15
14 Hypothesis two Sustained involuntary control 16
15 Hemodynamics as the strongest sustained control example 17
16 Bladder bowel and respiratory applications 18
17 Sexual and other involuntary responses 19
18 Bodily sensory interfaces and virtual experience 20
19 Medical application map 21
20 Consumer application map 22
21 Comparison with invasive brain interfaces 23
22 Research agenda and falsifiable predictions 24
23 Synthesis of the theoretical argument 25
Appendix How to read the evidence 26
References 27

1 Scope and method

The question this paper answers

Can implanted access to spinal circuits support future medical and consumer capabilities, including expert-informed motor training and sustained involuntary control, without violating known scientific principles? The intended conclusion is a set of conditional mechanisms and testable predictions. It is not a forecast of launch dates or an assertion that every proposed use will succeed.

A spinal cord–computer interface, abbreviated SCCI here, is a computational system with implanted access to spinal neural signals, spinal stimulation, or both. The term is used operationally because the literature more often describes spinal cord stimulation, spinal neuroprostheses, intraspinal recording, or brain–spine interfaces. An implanted electrode can be epidural, intradural on the cord surface, or penetrating. “Invasive” therefore does not mean that every electrode penetrates the spinal tissue.

Evidence selection

A targeted internet literature search examined primary studies in PubMed, publisher sites, and openly accessible manuscripts. Search themes included implanted spinal recording and decoding, patterned spinal stimulation, activity-dependent spinal plasticity, locomotion, upper-limb function, somatosensation, hemodynamics, bladder control, cough, and sexual function. Foundational anatomy and control theory were used to formulate the models. Publications available by the cutoff were eligible, including online-first articles; later publications were excluded. This is a narrative synthesis, not a registered systematic review or a pooled meta-analysis.

Direct support for SCCI applications comes from implanted spinal interventions. Noninvasive spinal stimulation, surface electromyography used alone as a “spinal interface,” peripheral nerve interfaces, and vagus stimulation are not counted as demonstrations of invasive SCCI. External movement or physiological sensors may form part of an otherwise invasive spinal system. Studies combining implanted cortical recording with spinal stimulation are explicitly labeled hybrid BCI–SCCI.

Interpretation rules

Human feasibility studies demonstrate that an effect can occur under particular conditions. Animal studies support mechanisms and recording strategies, with a separate inference required for humans. A model establishes what follows from its assumptions; it does not establish that a human nervous system satisfies those assumptions. Studies sharing participants are not independent replications. No sample sizes are pooled across the cited research programs.

Engineering and ethical barriers are outside the analysis. Biological resource requirements, missing neural pathways, non-identifiable signals, and limited control authority remain inside it: removing them would change the scientific question itself. The paper assumes future electrodes and computation can improve substantially, while avoiding an assumption of arbitrary access to every neuron or arbitrary rewriting of the brain.

2 What theoretical possibility means

An invention does not become impossible because it has not yet been demonstrated. Conversely, a statement that violates no obvious physical law is not yet a constructive explanation. For a research hypothesis, the useful middle ground is to describe an available causal pathway, the variables that must be measured and controlled, and the biological process that would produce the proposed outcome.

Classification	Meaning in this paper	Example
Demonstrated component	An experiment directly establishes a bounded effect.	Implanted stimulation evokes a tactile percept.
Supported extrapolation	Several relevant components work; their proposed extension is specified.	Adaptive bodily feedback for a virtual object.
Conditional hypothesis	A coherent mechanism exists, but one or more decisive biological assumptions remain open.	Expert-informed stimulation accelerates retained novice learning.
Missing causal access	The proposed access site lacks a demonstrated route for the requested control.	Native high-fidelity vision written through ordinary spinal sensory channels.
Contradiction under stated assumptions	A proposal conflicts with a defined physical or mathematical condition.	A decoder recovers task information absent from all its inputs.

These classifications are not probabilities. “Conditional hypothesis” can be scientifically valuable and technologically consequential. Confidence should attach to a particular proposition: high confidence that spinal stimulation can influence selected movement circuits does not imply high confidence that an expert-to-novice transfer system halves training time.

A constructive feasibility criterion

$$\text{Feasible task} = \text{relevant access} + \text{usable state information} + \text{adequate control} + \text{biological adaptation}$$

This expression is a checklist of necessary mechanisms, not a numerical law. For immediate assistance, adaptation may be unnecessary. For learning that remains after the implant is switched off, durable adaptation is essential. For a physiological regulator, adaptation may be undesirable if it destabilizes a previously calibrated controller.

A valid “on paper” claim can therefore be conditional: if task-relevant learner states are observable, stimulation can influence the required movement directions, and timed practice strengthens useful connections, then an expert-informed controller could accelerate that part of skill learning. The research program must determine whether these conditions hold together and whether the effect is meaningfully large.

3 The spinal cord as a processing and control system

The spinal cord is more than a passive cable. It contains motor neurons, sensory-processing networks, interneurons, reflex circuits, and autonomic pathways. It also carries ascending information to the brain and descending commands from it. Pattern-generating networks can organize aspects of rhythmic movement, but normal skilled behavior remains distributed across spinal circuits, brainstem, cerebellum, cortex, sensory organs, and the body. The popular "second brain" label more commonly refers to the enteric nervous system in the gut; it should not be treated as a technical description of a second general-purpose brain. [1–3]

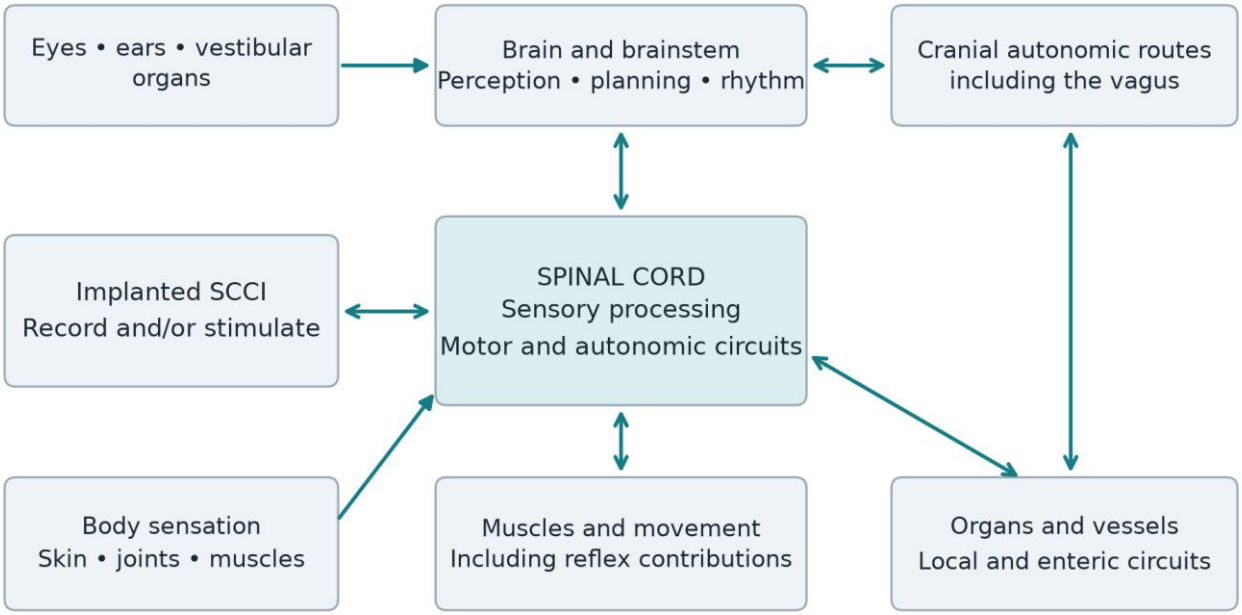


Figure 1. Functional access map. Solid arrows indicate major information or control routes; the diagram is schematic, not a complete tract atlas. Cranial sensory and brainstem pathways do not become directly accessible merely by adding a spinal electrode. Original diagram based on the anatomical organization in [1–3].

"Involuntary" describes how an action is experienced, not a single anatomical system. A spinal withdrawal reflex, respiratory rhythm generated largely in the brainstem, vascular sympathetic regulation, and local enteric motility have different circuits. Some outputs share spinal pathways, but this does not make every component independently controllable from the cord.

The relevant design question is consequently specific: which neural population carries a useful measurement or offers influence over the desired output? Cervical access is relevant to upper-limb circuits; lumbosacral access to lower-limb and pelvic circuits; thoracic access to selected trunk and sympathetic functions. Segmental neural anatomy must also be distinguished from vertebral level. [2,3]

4 Interface architectures and causal access

Architecture	What enters the computer	What it can change
Implanted spinal recording	Surface potentials, evoked responses, or intraspinal unit activity	A computer or prosthesis; recording alone does not stimulate tissue
Sensor-driven spinal stimulation	Movement, muscle activity, pressure, or other measured state	Recruitment and excitability of accessible spinal pathways
Spinal recording and stimulation	Spinal measurements plus optional external sensors	An adaptive spinal intervention; both read and write paths are implanted
Hybrid BCI–SCCI	Implanted cortical signals, optionally spinal and bodily feedback	Spinal stimulation linked to brain-derived intent

Conventional epidural stimulation frequently recruits dorsal-root afferents and influences motor circuits through their synapses. It therefore need not reproduce a motor neuron's natural spike train to assist movement. Computational and experimental work supports this afferent-mediated mechanism. Penetrating or circumferential interfaces provide different access, but none makes neural reading and neural writing equivalent operations. [4]

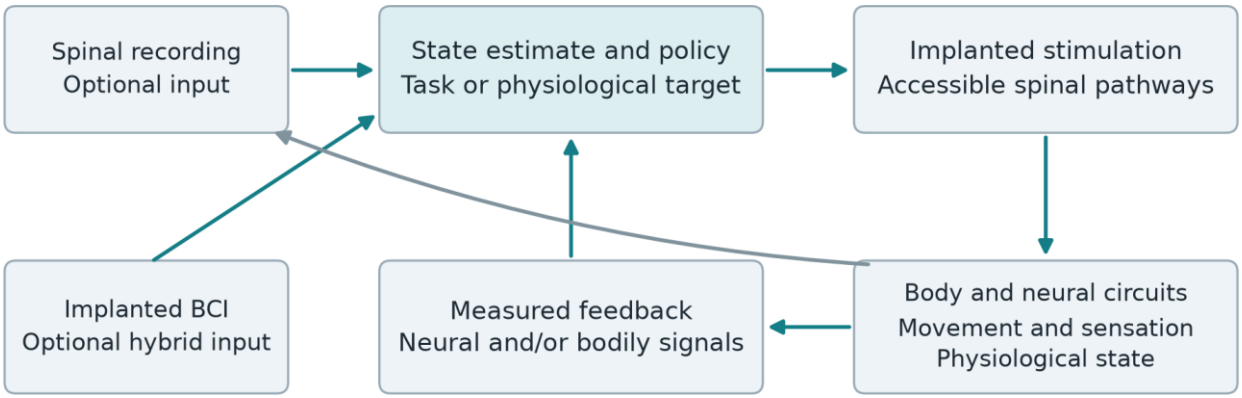


Figure 2. A general invasive SCCI loop. Neural recording is optional for some existing spinal neuroprostheses but essential to the proposed expert-recording hypothesis. The brain–spine route is a distinct hybrid architecture. Original conceptual diagram; examples in [5–9,14–18].

An electrode measures a weighted mixture of electrical sources. A stimulation electrode generates a field that excites structures according to their geometry and excitability. Replaying a recorded voltage through an electrode does not generally recreate the original source activity. A computational translation model, calibrated to the receiving nervous system, is the scientifically credible bridge between these operations.

5 Evidence that the spinal cord can be read

Implanted spinal recording provides an important foundation for the expert-template proposal. Fan and colleagues reported long-term intraspinal recording in mice and decoding of lower-limb movement from neural activity. Woodington and colleagues demonstrated circumferential recording and stimulation in animal preparations, including an acute injury-bridging proof of concept. These establish useful acquisition and interaction strategies, rather than a human expert-skill library. [5,6]

El Hadwe and colleagues extended surface circumferential recording to freely moving rodents in 2026. The study included movement-related spinal signals and decoding, with separate sensory and large-animal visceral experiments. The freely moving experiments involved five rats. Decoding a body trajectory establishes a statistical relationship with movement; it does not by itself identify a transferable command that caused that trajectory. [7]

Evidence	Supported inference	Inference still required
Chronic mouse intraspinal recording [5]	Spinal populations can contain decodable movement information.	Equivalent useful information from implanted human experts across tasks.
Circumferential animal interfaces [6,7]	Surface access can sample signals from different cord regions.	Stable decoding of task-relevant latent variables during diverse human behavior.
Human evoked-response feedback in pain treatment [18]	A spinal response to stimulation can regulate stimulation delivery.	That response reveals skill, intention, organ state, or subjective experience.

A recording can be informative without being a command

During movement, spinal signals can reflect descending drive, incoming sensory feedback, local processing, and stimulation artifacts. A decoder may predict joint motion well because it reads sensory consequences of movement. That is useful for state estimation, but copying those signals into another cord may not cause the motion. Identifying causal influence requires perturbation, timing analysis, and interventions, not prediction accuracy alone.

For an expert library, recordings should therefore be collected with task phase, posture, force, velocity, environmental context, and success or error labels. Neural features should be evaluated for information they add beyond ordinary kinematics or muscle recordings. The added value of the invasive recording itself is an empirical question; it should not be assumed simply because the signal is neural.

An especially strong experiment would show that a feature predicts a successful correction before the movement changes, and that a calibrated stimulation intervention aimed at the corresponding learner circuit improves that correction. Such a result would connect measurement, causal control, and training relevance.

6 Evidence that spinal stimulation changes useful outputs

Domain and study	Implanted evidence	Boundary of inference
Locomotion [8]	Three participants with clinically complete paralysis received task-specific epidural programs.	Clinical completeness is not proof of anatomical transection; support and training matter.
Brain–spine walking [9]	One participant used implanted cortical recording to control spinal stimulation.	Hybrid BCI–SCCI, not spinal-only decoding of intent.
Stroke upper limb [10]	Seven participants; mean 32% strength increase and 5.6 FMA-point improvement with stimulation on.	Feasibility cohort; finger benefits depended on residual connectivity in responsive participants.
Parkinson gait [11]	Animal work and one human case supported a spinal locomotor neuroprosthesis.	A complement to other treatment, not reversal of the disease.
Spinal muscular atrophy [12]	Three adults with type 3 SMA showed motor benefits during a four-week study.	Requires surviving circuits and muscles; does not imply replacement of lost motor neurons.
Sensorimotor feedback [15]	Three participants received above- and below-lesion stimulation for sensation and movement.	Intentional control used measured body movement; not a demonstration of spinal intent recording.

These results support a broad platform claim: accessible spinal networks can serve as an entry point for several useful interventions. They do not establish that one electrode configuration or one generic rhythm works across diseases. Distinct disorders alter descending input, sensory integrity, motor-neuron survival, and the balance of excitation and inhibition.

For consumer hypotheses, these clinical studies provide evidence of causal access to human circuits. The additional inference is that modulating an intact system improves performance or learning. Restoration after a lesion and enhancement above an intact baseline are biologically different problems. A healthy circuit may already operate near an effective solution, and extra stimulation may interfere with it.

The relevant opportunity is therefore targeted assistance in a measurable task direction: for example, facilitating a timely extension, stabilizing a stance phase, or supplying informative bodily feedback. The research target should be a defined improvement in coordination or learning, rather than an undifferentiated increase in “human performance.”

7 Physics and information constraints

Electrical fields provide a physical mechanism

$$\nabla \cdot (\sigma \nabla \varphi) = 0 \quad \text{in source-free tissue under a quasistatic approximation}$$

Here φ is electric potential and σ is tissue conductivity, which may be direction-dependent. Electrode currents enter through boundary conditions. This volume-conductor approximation connects an imposed current to a field in tissue. Axon excitation then depends on membrane dynamics, orientation, and spatial variation of the extracellular potential. It is not determined solely by whether the imposed waveform resembles a recorded waveform. Computational spinal models make this field-to-neuron connection explicit. [4]

Ignoring manufacturing limitations can permit denser arrays and better field shaping in a hypothetical design. It does not remove the fact that multiple neural structures share a conductive medium. Perfect selectivity must be stated as an extra assumption if a hypothesis depends on it. Different electrode positions and neural access strategies can change the available control directions.

Formento and colleagues showed why preserving proprioceptive information matters during spinal stimulation. Interactions between imposed stimulation and naturally traveling sensory signals can disrupt useful feedback. This is a concrete biological reason to optimize timing and pathway recruitment rather than assume that more stimulation always produces better control. [24]

Information cannot be reconstructed from an absent input

$$I(S ; D(Y)) \leq I(S ; Y)$$

For a target variable S, recorded data Y, and a decoder D that uses only Y, the data-processing inequality says that decoding cannot add information about S beyond that present in Y. This does not forbid useful inference from partial signals. It says that better computation alone cannot identify distinctions the measurement never contains. Additional sensors or informative context change the input and can change the result. [26]

For skill transfer, a spinal signal may contain execution timing without containing the expert's strategic choice or visual prediction. A system can still accelerate the execution component if it has enough relevant information. The target need not be the entirety of the expert's knowledge.

Timing and energy remain biological requirements

The total delay includes sensing, computation, conduction, synapses, and muscle activation. A useful controller must act within the timescale of the correction it is meant to improve. A phase-aware predictive policy could compensate for some delay. It cannot make neural conduction instantaneous.

$$\text{Mechanical power} = \text{joint torque} \times \text{angular velocity}$$

Stimulation supplies a control signal; contracting muscle supplies most of the mechanical work. Oxygen delivery, fuel, force–velocity relationships, fatigue, and tissue recovery still matter. A future SCCI could improve coordination or distribute effort more effectively, but prolonged activation is not unlimited endurance.

8 A mathematical model of task control

Let x describe relevant neural and bodily state, u the stimulation command, d disturbances, and y measured outputs. A general model is:

$$dx/dt = f(x, u, d) \qquad y = h(x) + \text{noise}$$

The equations are a proposed modeling framework, not a fitted human spinal model. A local linear approximation around a particular operating state gives dx/dt = Ax + Bu and y = Cx. Matrix B describes the state directions accessible to stimulation; C describes what the sensors reveal. Standard control theory provides tests of reachability and observability for such specified models. [25]

$$R = [B \;\; AB \;\; A^2B \;\; \ldots \;\; A^{n-1}B] \qquad O = [C; CA; CA^2; \ldots; CA^{n-1}]$$

Full rank n would imply complete local reachability or observability in this ideal linear model. No cited study establishes these rank conditions for an entire human spinal cord. Fortunately, a useful product normally needs control of a small task output, such as grip timing or pressure, rather than every neural state. Relevant unstable modes must be controllable and detectable; task outputs must be identifiable and reachable within allowable inputs.

An explicit counterexample to universal replay

$$x_1' = u \qquad x_2' = u \qquad \Rightarrow \qquad (x_1 - x_2)' = 0$$

If one effective input always changes two outputs together, it cannot independently set both. Unlimited processor speed does not alter that dependency. Adding another genuinely independent neural control pathway could alter it. This simple example explains why observing one organ response cannot establish arbitrary independent control of all organs.

A constructive condition for expert trajectory assistance

$$\dot{r}E(t) - fL(rE(t), 0, d) \in \text{range } BL(rE(t))$$

For a simplified control-affine learner model, this condition says that the correction needed to follow a suitably mapped expert reference rE must lie in the learner's accessible control directions. BL is the learner's stimulation influence matrix, and fL its unstimulated dynamics. If this condition holds locally, an input solving the inverse problem can in principle generate the correction, subject to input bounds and state estimation. If it fails, the reference must be adapted or additional access is required. This is a conditional mathematical argument, not an empirical claim that all expert trajectories are reachable.

The model makes the important distinction precise: future hardware can enlarge the accessible set, but a useful hypothesis should identify the intended set instead of quietly assuming arbitrary control.

9 Hypothesis one Expert informed skill acceleration

Hypothesis H1: an invasive SCCI that extracts task-relevant coordination features from expert spinal recordings, translates them into a learner-specific stimulation policy, and applies that policy during active practice can reduce the training required to reach a predefined level of retained, unassisted motor performance.

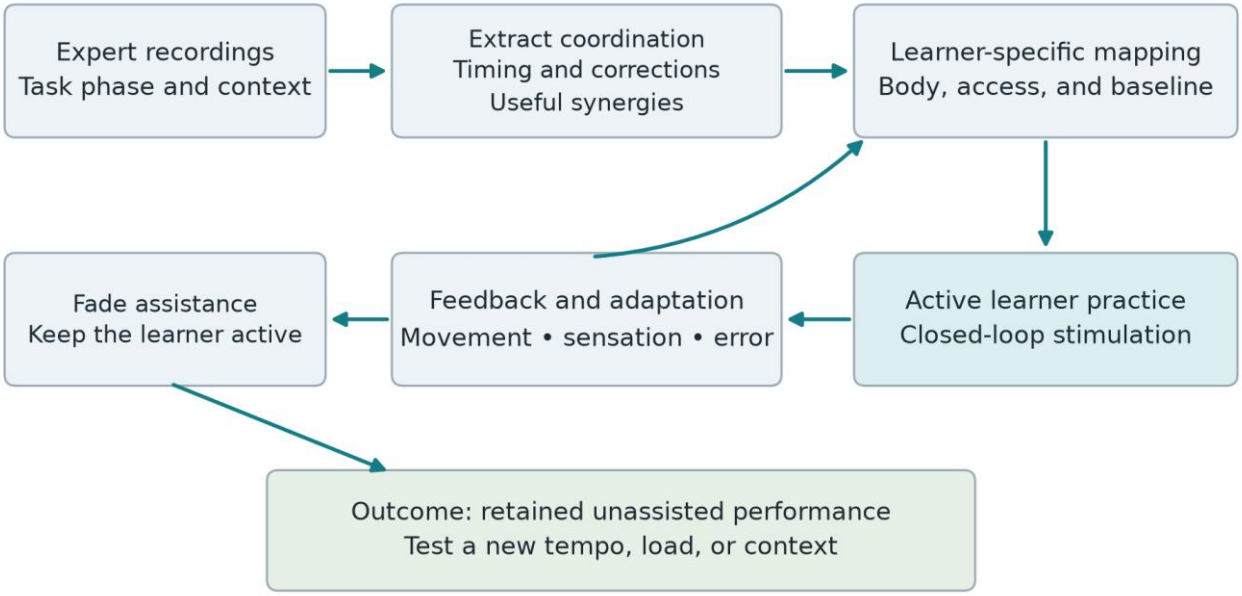


Figure 3. Proposed expert-to-learner training architecture. The transfer object is a context-dependent coordination policy. Learner calibration, active attempts, sensory consequences, and off-stimulation assessment are essential parts of the hypothesis. Original diagram; component evidence in [5–7,20–23].

The expert contributes candidate patterns of timing, recruitment, intermuscular coordination, and response to perturbations. The learner contributes a different nervous system, body, starting skill, and feedback history. A mapping must align task phase and functional output, rather than assume that matching electrode locations or voltage traces matches neural computation.

$$u_L(t) = \pi_\theta[z_E(\text{phase}, \text{context}), \hat{x}_L(t), \text{goal}]$$

Here z_E is a learned representation of the expert's task-related spinal activity; $\hat{x}_L$ is the estimated learner state; and π_θ is the individually calibrated policy. The equation defines a research architecture. It does not imply that the expert representation is uniquely identifiable, that it includes all of expertise, or that every representation has an effective stimulation inverse.

The outcome could be revolutionary even if it is restricted to part of a skill. Faster acquisition of a reliable grasp, a stable stepping pattern, or a difficult movement sequence could have substantial value. Strategy, perceptual judgment, and conceptual understanding would still be learned through their own relevant pathways.

10 Biological support for the learning hypothesis

Component one Coordinated outputs can have lower dimensional structure

Mussa-Ivaldi, Giszter, and Bizzi showed that stimulation-evoked motor fields in frogs could combine in approximately additive ways. This supports the idea that useful motor outputs may be constructed from reusable coordination components. It does not establish a universal library of human motor primitives, but motivates testing whether expert information can be represented in a smaller task-specific space. [20]

Component two Timing can change connection strength

Nishimura and colleagues linked cortical spikes to intraspinal stimulation in two monkeys. The timing of this implanted hybrid interface altered corticospinal connection strength; some changes lasted for days, although many reverted. This supplies direct evidence that appropriately timed neural activity and spinal stimulation can alter connections in vivo. It is not a demonstration of spinal-only expert transfer. [21]

Component three Activity linked stimulation can improve lasting function

McPherson and colleagues used muscle-activity-triggered intraspinal stimulation with training after cervical injury in rats. The activity-linked group recovered substantially more skilled reaching than open-loop stimulation or retraining alone, with benefits persisting after stimulation stopped. The reported final recovery was roughly twice that of the comparison groups. This is a recovery endpoint in injured animals, not evidence of a twofold learning-speed increase in healthy people. [22]

Kathe and colleagues combined human epidural-stimulation rehabilitation observations with mouse circuit analysis to identify a neuronal population important for recovery in that model. The work strengthens the argument that stimulation and practice can reorganize relevant circuits. It does not show that an expert's learned policy is stored in a single spinal population. [23]

The missing conjunction

Together, these findings support recording, controllable movement components, and activity-dependent adaptation. H1 requires their conjunction: a representation learned from expert spinal activity must causally improve the novice's practice in a way that produces retained learning. The cited studies do not directly test that conjunction. The hypothesis is stronger than pure speculation because its component mechanisms exist, while its claimed benefit remains open.

A possible mechanism is better timed practice: stimulation makes useful movements and sensory consequences occur while the learner generates an intention or correction. Repeated successful pairings could strengthen useful pathways or improve internal predictions. Another possible mechanism is informative sensory cueing that helps the learner recognize a phase or error. Either mechanism could work without copying the expert's synapses or transferring declarative knowledge.

11 What a learning multiplier would mean

Acceleration A = training time in control / training time with SCCI

Both times must refer to the same predeclared proficiency threshold, measured without stimulation after a retention interval. A = 1.5 means reaching that threshold in two-thirds of the control training time; A = 2 means half the time. Better performance while stimulation is on is a separate assistive outcome.

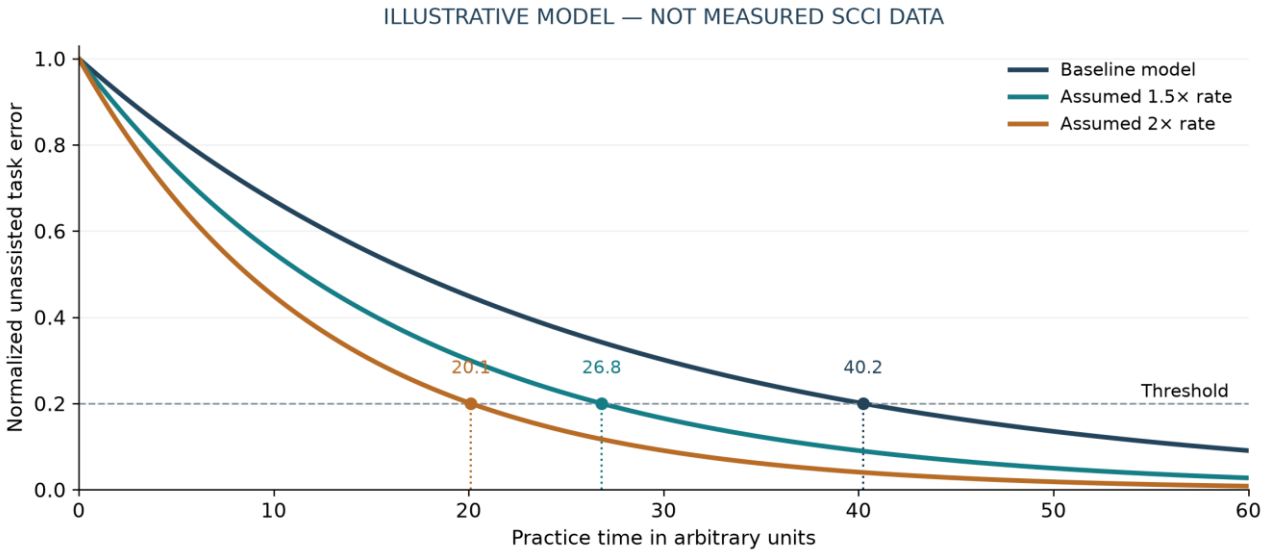


Figure 4. Illustrative learning curves, not experimental data. The assumed error model is E(t) = exp(−kt), with k = 0.04, 0.06, and 0.08 per arbitrary practice unit. A threshold E = 0.2 is reached at approximately 40.2, 26.8, and 20.1 units. Equal initial error and equal asymptotic error are assumed. The plot explains the metric; it does not estimate an SCCI effect.

For the more general model $E(t) = E_\infty + (E_0 - E_\infty) \exp(-kt)$, the threshold time depends on the starting error E_0, the achievable asymptote E_∞, the learning-rate parameter k, and the chosen threshold. A change in final achievable performance can therefore look different from a change in learning rate. A single endpoint cannot uniquely identify both.

Nothing in known physics singles out 1.5 or 2 as impossible multipliers for a bounded training task. Nothing in the current spinal literature establishes either as a general human expectation. Estimating a multiplier requires data on the actual task, participants, baseline training, and retained outcomes. Some tasks may improve substantially; some may show little effect or impaired learning.

Total user time should also be reported. A system that halves practice time but requires extensive individual calibration may not halve the time from first use to proficiency. This is an accounting distinction even when engineering costs are ignored.

12 Candidate learning applications

Application	Hypothesized trainable component	What would count as success
Medical rehabilitation	Reaching, grasp release, stance-to-swing timing, coordinated trunk control	Faster recovery to a retained functional threshold with stimulation off
Sports and dance	Repeatable timing, balance corrections, sequencing, reduced unnecessary co-contraction	Improved retained coordination at matched speed and task difficulty
Musical performance	A limited finger or limb sequence and graded force timing	Unassisted sequence accuracy and transfer to new tempo or context
Industrial or surgical simulation	Stable limb trajectories or precise force transitions	Reduced training to a validated motor subtask criterion
Embodied games and virtual training	Learning a mapping between bodily cues and task actions	Retained control of the mapping beyond device-assisted trials

These are proposed applications, not demonstrated SCCI products. The most plausible initial targets have a measurable motor component, repeated trials, a clear phase structure, and a tractable relationship between spinal access and output. Rehabilitation has the closest direct experimental support. Healthy consumer enhancement requires its own tests.

Example A difficult hand sequence

An expert performs a short finger sequence at several speeds. Spinal recordings are analyzed alongside joint motion and force. The useful target may be release timing or suppression of excessive co-contraction, rather than every spike associated with the performance. The novice's stimulation map is calibrated to identify which accessible inputs influence that target. During practice, the controller offers phase-appropriate assistance or bodily cues, then gradually withdraws them. The hypothesis is confirmed only if the novice later performs the sequence better without assistance and transfers it to an untrained tempo.

Example Faster relearning after injury

A person recovering a grasp may have sufficient surviving circuitry to benefit from stimulation aligned with attempts and sensory feedback. An expert or pre-injury template could propose useful timing, but an individualized target may outperform a healthy expert template when anatomy or available movement differs. The medically relevant aim is a better achievable action for that person, not forced imitation of a body with different capabilities.

The paper therefore predicts a conditional advantage for expert recordings: they help when they supply task-relevant information beyond motion capture, muscle activity, and task models. If a simpler reference performs equally well, the learning intervention may still work, but the specific value of invasive expert recording would be unproven.

13 How to test expert informed learning

Separate assistance from learning

The primary endpoint should be retained unassisted performance after a predefined interval, with a later follow-up and a transfer test. Stimulation-on performance measures immediate assistance. Stimulation-off performance during training can reveal emerging learning, while delayed tests distinguish a short-lived state change from durable skill.

Comparison	Question answered
Matched practice with sham or appropriate control stimulation	Does the intervention add to practice and expectation?
Learner-specific closed-loop versus time-yoked stimulation	Does contingency on the learner's actual state matter?
Expert neural template versus kinematic or muscle template	Do implanted expert recordings contribute unique value?
Expert-informed versus individually optimized policy	Is expert transfer better than learning a policy directly for the novice?
Expert template versus nonexpert or phase-scrambled template	Do expert structure and temporal alignment matter?

A staged program can test these questions sequentially rather than attempt an underpowered trial with every arm. Practice amount, instructions, feedback availability, motivation, and task difficulty should be matched or explicitly modeled. Analyses should account for baseline ability and repeated measures. Sample size must follow pilot variability and a prespecified effect, rather than being invented from the desired multiplier.

Predictions that could support or weaken H1

H1 predicts a benefit that depends on appropriate task phase, learner calibration, and active engagement. A misplaced phase or a poorly matched template should reduce the effect. A successful model should generalize to withheld trials and tolerate moderate changes in context. A benefit that disappears immediately when stimulation ends supports assistance but not the retained-learning claim.

Excessive guidance may also reduce the learner's opportunity to detect and correct errors. Behavioral motor-learning experiments support testing this possibility, although they are not invasive SCCI evidence. A policy that fades assistance and preserves active error correction is therefore a hypothesis worth comparing with continuous guidance. [27]

A rigorous null result would still be informative. If neural templates add no value beyond ordinary movement references, the record-from-an-expert component needs revision. If stimulation reliably improves performance but not retention, the product belongs in the assistive category. If gains transfer only to an identical repeated sequence, the claim should remain sequence-specific.

14 Hypothesis two Sustained involuntary control

Hypothesis H2: an invasive SCCI can extend useful control over a selected physiological state or action by estimating its current state and applying stimulation that maintains a target, repeats an output, or changes a transition at the appropriate time. Recordings of successful natural behavior can inform the target policy, but the control action must be adapted to the recipient.

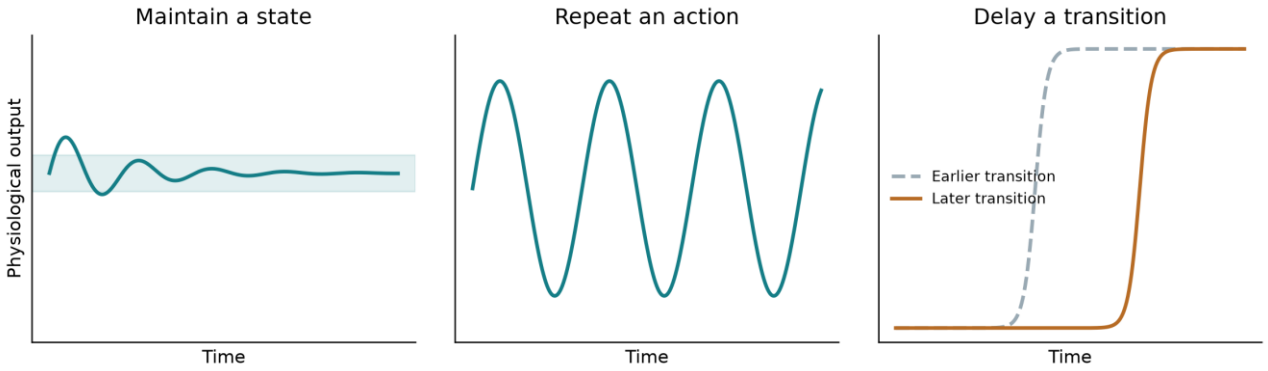


Figure 5. Three distinct meanings of "longer control." Holding a pressure range, repeating a motor pattern, and delaying a state transition require different models and measurements. The traces are conceptual and have no empirical units. Original diagram.

The first target is a state, such as maintaining blood pressure in a useful range while posture changes. The second is a repeated action, such as producing effective coughs when required. The third is a transition, such as moving between bladder storage and voiding or modifying the timing of an ejaculation-related reflex. Evidence for one category does not automatically establish the others.

$$\text{For a maintained state: } f(x^*, u^*, d) = 0$$

The target x^* must admit an achievable input u^* and remain stable under disturbances. For a rhythm, the target is a repeatable trajectory or cycle rather than an equilibrium. For a delayed transition, the controller must influence the relevant transition mechanism while preserving the desired preceding state. These conditions make the "record and imitate for longer" idea concrete.

Open-loop replay may help in a stable, repeatable situation. Feedback is more compelling for long-duration control because posture, fatigue, bladder filling, vascular tone, and background neural drive change over time. A recorded successful episode is a reference for a controller, rather than a guarantee that repetition will reproduce the same bodily result.

15 Hemodynamics as the strongest sustained control example

Phillips and colleagues reported an implantable spinal system for hemodynamic instability after spinal cord injury in 2025. Across a staged program involving 14 intervention participants, targeted lower-thoracic stimulation and feedback-supported operation demonstrated clinically relevant pressure regulation. The study's much larger survey population describes disease burden and must not be confused with its implant sample; individual experimental stages used smaller subsets. [13]

This is an unusually direct foundation for H2. The controlled variable is physiological, changes with disturbances, and can be measured repeatedly. The proposed future extension is a more individualized regulator that adapts to posture, activity, and context. That extension does not require assuming that the user consciously knows how to generate the appropriate sympathetic pattern.

A minimal constructive model

$$dp/dt = -a(p - p_0) + b\,u + d(t) \qquad \text{with } a > 0 \text{ and } b > 0$$

In this illustrative local model, p is pressure, p_0 a baseline, u an effective spinal input, and d a disturbance. Around an attainable operating point, proportional feedback $u = u^* + K(p^* - p)$, with $K > 0$, gives local error dynamics $e' = -(a + bK)e$ for a fixed disturbance and an unsaturated actuator. The negative coefficient explains why feedback can stabilize the state. This is a mathematical example, not a physiological fit or a stimulation prescription.

Real cardiovascular control has delays, nonlinearities, changing gain, saturation, and interactions with the native baroreflex. If stimulation primarily raises pressure, the system's authority is asymmetric: reducing stimulation can remove that added drive, but does not imply an independent active mechanism for lowering arbitrarily high pressure. A future controller must operate within its attainable range.

Medical and consumer implications

The strongest medical use is compensation for impaired autonomic regulation. A consumer extension might target selected episodes of orthostatic intolerance or assist posture-related physiological transitions in a population shown to benefit. General athletic enhancement or arbitrary control of heart rate does not follow from the pressure result. Different outputs may require different pathways and measurements.

The logical lesson is positive and specific: an involuntary variable can be brought into a computer-mediated feedback loop through spinal access. This is already more than a speculative analogy. Expanding the set of controllable variables is a research direction, with each added variable requiring evidence of access and adequate control authority.

16 Bladder bowel and respiratory applications

Bladder state transitions

Hubscher and colleagues reported a 2026 pilot in three people with chronic spinal cord injury using multiple epidural programs for bladder-related functions. The findings support the possibility of adapting stimulation to storage and voiding requirements over extended use. The small uncontrolled study does not establish universal restoration or an unlimited ability to postpone urination. [14]

A future controller could combine bladder-state estimates, pressure or other physiological measurements, and the user's intention to schedule transitions. A successful storage intervention must preserve an appropriate relation among capacity, pressure, and outlet behavior. Merely preventing emptying is not equivalent to functional continence. Similarly, contraction without coordinated outlet relaxation may not produce effective voiding.

For a consumer scenario, longer comfortable control during travel is a conceivable bounded target. Its benefit must be defined as an extension within the organ's functional operating range. The finite filling process remains a biological constraint. Bowel scheduling is also plausible in selected contexts, but its more distributed enteric and autonomic control warrants a separate model; bladder success alone does not prove it. [3,14]

Cough and respiratory motor assistance

DiMarco and colleagues demonstrated implanted lower-thoracic stimulation for expiratory muscle activation and cough in people with spinal cord injury. This provides direct evidence that a useful respiratory-related motor output can be recruited through spinal access. It supports future feedback-guided cough assistance and secretion-clearance systems. [16]

Sustaining breathing is a broader problem. It requires appropriate inspiratory and expiratory mechanics, airway patency, gas exchange, and regulation according to oxygen and carbon dioxide. Brainstem rhythm generation is distinct from spinal access to respiratory muscles. A cough result therefore supports a specific output, while a complete respiratory controller requires additional sensing and validated control of the necessary muscle groups. [2,3]

Why normal physiology is not a single replay file

Storage, voiding, cough, and ventilation interact with changing loads and organ states. A pattern that works at one lung volume or one bladder filling level may work differently later. The biologically credible "imitate" strategy is to learn a policy conditioned on these states, with the option to stop, switch phase, or reduce input when the target has been reached.

Restoring a missing output and prolonging an existing output should be reported separately. Both may be useful, but they test different claims about the interface.

17 Sexual and other involuntary responses

Separate the physiological endpoints

Erection, genital arousal, emission, expulsion, orgasm, and subjective pleasure are related but distinct. Their spinal, autonomic, peripheral, endocrine, and supraspinal contributions differ. A useful hypothesis names which endpoint is being controlled and which measurements would establish that control.

In a two-case report, Rybka and colleagues described epidural stimulation facilitating ejaculatory responses in men with spinal cord injury, including responses associated with penile vibratory stimulation. Shackleton and colleagues reported questionnaire changes in sexual function in three women receiving epidural stimulation after spinal injury. These small studies support physiological influence and motivate further investigation; they do not measure controllable prolongation of arousal or delayed ejaculation. [28,29]

In anesthetized rats, Borgdorff and colleagues evoked ejaculation through microstimulation of lumbar spinothalamic neurons. This provides a causal animal example of access to a coordinated sexual motor sequence. Translating that access into selective human prolongation, inhibition, or subjective experience is a separate step. [30]

A constructive prolongation hypothesis

A controller could, in principle, estimate a selected physiological state, maintain the drive needed to support it, and separately regulate a transition that would end or change it. For prolonged erection or arousal-related bodily responses, the vascular state and opposing autonomic influences must be controllable. For delayed ejaculation, the decisive requirement is access that can postpone or inhibit the relevant transition without simply destroying the desired preceding response.

Recording a longer successful episode could supply a target trajectory. However, stimulation that triggers ejaculation supplies evidence of excitatory access, not necessarily the inhibitory or selective access needed to delay it. More selective spinal access could change that conclusion, but the mechanism must be identified. Maintaining a peripheral response also does not establish sustained orgasm or an arbitrarily prolonged subjective experience.

Other candidate states

Targeted regulation of muscle tone, reflex gain, or selected autonomic patterns is mechanistically plausible when an accessible pathway and useful feedback variable can be identified. Sweating, thermoregulation, gastrointestinal activity, and metabolic regulation involve additional distributed and sometimes endocrine processes. They are research candidates, not consequences of a universal spinal control principle.

H2 is therefore best framed as a family of organ- and endpoint-specific controllers. The record-and-imitate idea can contribute to each, but the necessary causal experiment is different for each physiological endpoint.

18 Bodily sensory interfaces and virtual experience

Chandrasekaran and colleagues showed that lateral cervical epidural stimulation could evoke sensations referred to the missing limb in four upper-limb amputees. Nanivadekar and colleagues then demonstrated closed-loop sensory discrimination in two participants using implanted spinal stimulation linked to interaction with objects. These studies support a concrete route to useful artificial bodily feedback. They do not establish a complete natural somatosensory replacement. [17,19]

Calvert and colleagues demonstrated combined above-lesion sensory and below-lesion motor stimulation in three participants with motor-complete spinal injury in 2026. This is relevant to a future integrated bodily interface because movement and an informative sensation can be coordinated through implanted spinal access. The interface used body-movement-derived control rather than spinal neural decoding of intention. [15]

Future use	Plausible spinal contribution	Additional requirement
Prosthetic touch	Contact, force-related cues, or object-state information	Reliable percept mapping and integration with the prosthesis
Virtual bodily feedback	Touch-like cues and movement-related bodily signals	Validation of perceptual quality, localization, and interaction
Teleoperation	Remote sensor data mapped to interpretable bodily cues	Learnable mapping and appropriate feedback timing
Movement guidance	Phase or error cues delivered through bodily sensation	Evidence that cues improve retained performance
Full-sensory virtual reality	Some bodily channels could contribute	Separate causal access to native visual, auditory, and vestibular representations

For consumer products, a limited sensory vocabulary can still be useful. A user could learn that a particular percept indicates contact force, balance error, or a remote event. Sensory substitution may convey information without recreating the normal sensation associated with that information.

Complete full-dive virtual reality through SCCI alone is a much stronger proposition. Ordinary spinal access is not a direct gateway to retinal, auditory, and vestibular afferent pathways. Indirect brain effects or extensive learned remapping are not logically excluded, but they require additional biological hypotheses about how arbitrary, high-fidelity multimodal experiences would be written. The evidence for bodily feedback cannot carry that entire inference.

A hybrid immersive system could combine spinal bodily feedback with conventional sensory displays or other invasive neural sites. Its theoretical case is different because it explicitly adds the missing input channels.

19 Medical application map

Application family	Evidence position	Credible future direction
SCI movement and rehabilitation	Direct human stimulation and hybrid-interface evidence [8,9,23]	Adaptive movement assistance with feedback and individualized training
Stroke upper-limb function	Human feasibility evidence [10]	Daily assistance; testing retained training benefits and selection criteria
Parkinson locomotion	Animal evidence plus a human case [11]	Context-sensitive gait support integrated with other treatment
Motor-neuron disease symptoms	Small SMA feasibility study [12]	Improved recruitment of surviving circuitry in selected conditions
Autonomic pressure regulation	Human implant and feedback evidence [13]	Disturbance-aware physiological stabilization
Bladder and pelvic function	Small human studies [14,28,29]	Endpoint-specific storage, voiding, and sexual-function programs
Cough assistance	Human implanted expiratory activation [16]	Sensor-guided cough timing and repeated useful output
Sensory restoration	Small human percept and task studies [15,17,19]	Integrated prosthetic and bodily feedback
Selected chronic pain	Randomized implanted-device evidence with important comparator dependence [18,31]	Individualized neural-response-guided stimulation

Pain treatment requires particular care in interpreting endpoints. The EVOKE trial compared evoked-response-controlled stimulation with fixed-output stimulation in 134 randomized participants. Its recorded signal reflects neural recruitment rather than a direct readout of subjective pain. In a different indication and stimulation mode, Hara and colleagues found no disability advantage of burst stimulation over placebo. These results support indication-specific analysis rather than a universal spinal “pain off” switch. [18,31]

The common future direction across these applications is adaptive specificity: select a function, identify the useful circuit access, measure relevant state, and adjust stimulation to context. The same computational framework can apply across functions, while the physiology, electrodes, measurements, and endpoints remain function-specific.

20 Consumer application map

Consumer concept	Mechanistic case	Confidence boundary
Motor training coach	Expert-informed timing and learner-calibrated assistance during active practice	Plausible H1; exact speedup and benefit in healthy users unestablished
Bodily feedback for games and virtual environments	Spinally evoked, task-linked sensory cues	Human component evidence; breadth and naturalness require testing
Remote embodied interaction	Encode external contact or motion measurements as bodily information	Plausible sensory substitution; not transfer of another person's complete experience
Coordination assistance	Phase-specific influence on movement and reflex contributions	Causal motor access supported; enhancement above intact baseline uncertain
Selected physiological regulation	Closed-loop maintenance of an accessible variable	Strongest precedent in impaired pressure regulation; each healthy-use target needs evidence
Pelvic timing or sexual-function support	Control of a named state or transition	Small component studies; selective prolongation is a separate hypothesis

These categories describe scientific possibilities, not a recommendation that healthy people obtain implants. Ethical and market assessments are outside the paper's scope. The distinction between clinical restoration and consumer enhancement is retained because it affects biological inference, even when all adoption barriers are ignored.

Why consumer potential need not mean superhuman performance

A valuable consumer system could reduce variability, improve the consistency of a difficult movement, provide a new sensory channel, or shorten training for a bounded task. Such benefits do not require exceeding every biological limit. They do require a baseline with room for improvement and evidence that the intervention does not simply move the performance cost into another dimension, such as accuracy, attention, or fatigue.

An intact nervous system can also resist or compensate for imposed patterns. The optimal policy may therefore be subtle cueing or assistance only when an error is developing. Whether the best solution stimulates motor-related circuitry, sensory pathways, or both should be determined by the task and the learner's response.

The most defensible consumer forecast is a set of narrow, individualized capabilities that may later combine. Calling the platform general-purpose should refer to reusable computational infrastructure, not to evidence that every bodily or mental function is controllable through the spinal cord.

21 Comparison with invasive brain interfaces

Capability	Invasive SCCI	Invasive BCI or hybrid
Movement execution	Direct access to spinal sensorimotor pathways and bodily outputs	Brain-derived intent can command spinal stimulation or prosthetic effectors
Bodily feedback	Can engage ascending somatosensory routes at spinal levels	Other central sites can access different sensory representations
Speech and language output	No demonstrated general spinal route to linguistic content	Intracortical speech decoding has direct human evidence [32]
Motor learning	Potential to shape practice, sensory consequences, and spinal plasticity	Can additionally target or read selected supraspinal learning-related processes
Autonomic regulation	Access to selected spinal autonomic pathways	Different central targets may influence other aspects; access remains specific
Full-sensory immersion	Spinal access alone lacks direct coverage of all required modalities	Distributed access could broaden channels; complete arbitrary immersion remains a separate hypothesis

Willett and colleagues demonstrated an implanted speech neuroprosthesis that decoded attempted speech in a participant with ALS. That result illustrates why interface location matters: language-related motor representations can be available at cortical sites without being recoverable from ordinary spinal recordings. It does not imply unrestricted decoding of private thought. [32]

The advantage of SCCI is proximity to selected bodily circuits. The advantage of a BCI depends on which brain region is accessed and what representation it exposes. A brain–spine system can join intent-related signals with spinal effectors, as demonstrated for walking. Its success should not be described as a spinal-only capability. [9]

Revisiting instant skill transfer

A complete expert skill includes sensory discrimination, predictions, strategy, action selection, and learned bodily control. No evidence here establishes that these distributed states can be copied through a spinal interface. The stronger constructive research target is acceleration of a specified component through better practice and plasticity. This avoids both an unsupported total-download claim and an unnecessary declaration that all neural learning enhancement is impossible.

Even in an idealized future, “instant” would need a defined timescale. Neural signaling and changes in biological state take finite time. The relevant scientific question is how much the required learning trajectory can be shortened, which components can be influenced, and what remains after the external support ends.

22 Research agenda and falsifiable predictions

Hypothesis	Decisive experiment	Result that would limit the claim
H1 Expert-informed learning	Matched training with delayed off-stimulation retention and transfer	Benefit exists only while stimulation is active
H1a Neural expert data add value	Compare expert spinal features with non-neural references under matched control	No incremental benefit from neural recordings
H1b Timing drives adaptation	Compare contingent, phase-shifted, and time-yoked interventions	Effect is insensitive to the predicted timing relationship
H2 Maintained state	Apply disturbances while regulating a measured physiological output	Target cannot be stabilized within accessible input directions
H2a Extended duration	Compare duration of functional control at matched physiological conditions	Longer signal delivery does not extend useful output
H2b Delayed transition	Measure a named transition while maintaining its preceding desired state	Stimulation only triggers or suppresses the entire response

A productive sequence begins with a narrow task and a model whose assumptions can be measured. Map accessible outputs, estimate state and delays, test immediate closed-loop control, and only then test retained learning or longer-duration regulation. Demonstrating a useful input–output relationship is a prerequisite for a transfer experiment, not a substitute for it.

Expert-to-learner studies should compare within-person and between-person templates. If a person's own best trials are more effective than an external expert's recordings, the platform might be better understood as self-calibrating training. If expert templates help only after transformation into task variables, the transferable object is that representation rather than the raw neural waveform.

Physiological-control studies should distinguish sensing the target variable directly from inferring it from spinal signals. A neural estimator may be useful, but an external pressure or movement measurement can be scientifically sufficient for a spinal stimulation controller. The claim should identify which sensors were actually used.

All learning and duration estimates should include uncertainty intervals and individual responses. Aggregate averages may conceal a subgroup with strong benefit, no benefit, or interference. Models should be tested on withheld sessions and conditions, with performance assessed on the variable the product claims to improve.

23 Synthesis of the theoretical argument

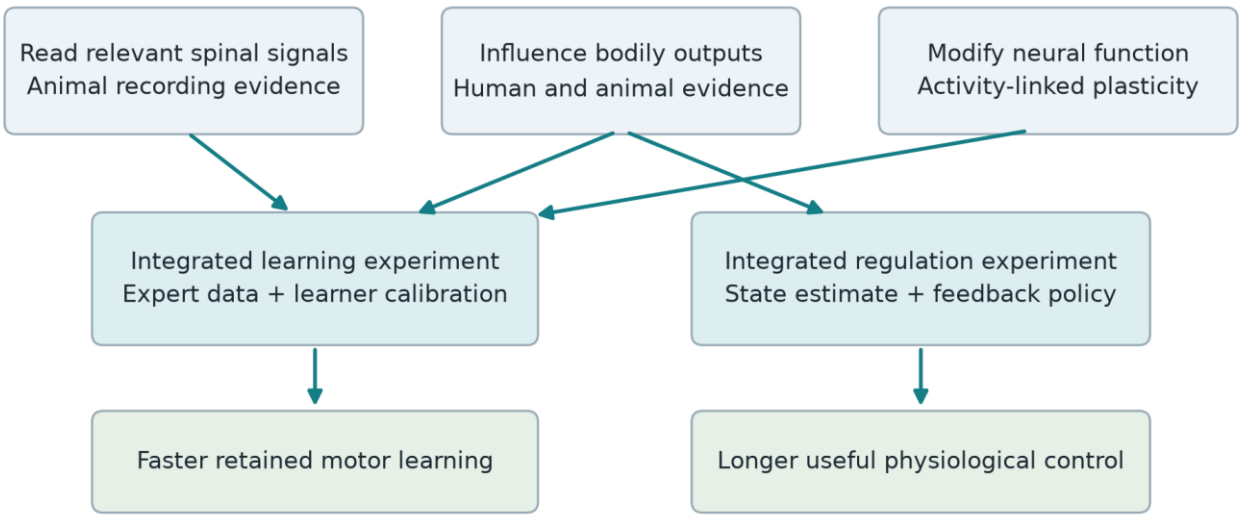


Figure 6. Evidence-to-hypothesis map. Arrows indicate logical dependencies, not proof that the final application works. Separate component demonstrations motivate integrated experiments. Original synthesis of the evidence cited throughout the paper.

The expert-training hypothesis has a coherent route through known biology. Spinal recordings can contain movement-related information; spinal inputs can alter movement and sensation; and the relationship between activity, stimulation, and subsequent function can depend on timing. Those premises justify serious investigation of faster retained learning. The unresolved step is whether expert-derived neural information improves an individually calibrated training policy sufficiently to produce a meaningful advantage.

The sustained-control hypothesis has direct human precedent for a selected physiological variable. Its expansion to other functions is plausible when the relevant state can be estimated and the spinal access can exert the needed influence. Maintaining a state, repeating an action, and postponing a transition remain distinct scientific objectives.

The defensible conclusion is therefore affirmative but bounded: invasive SCCI is a plausible platform for substantial medical and consumer capabilities under known science, including some forms of learning acceleration and longer control over selected bodily functions. A 1.5-fold or twofold learning benefit is a testable target, not a result derivable from physics alone. A future physiological controller is a collection of specific causal capabilities, not evidence of universal body control.

Historical lack of demonstration does not refute these proposals. Their strength comes from identifying what would make them work, what evidence already supports those mechanisms, and which experiment could overturn the remaining assumption.

Appendix How to read the evidence

Term	Operational meaning
Invasive	At least one neural interface is surgically implanted; epidural electrodes qualify.
Closed loop	The intervention changes in response to measurements, not simply a preset schedule.
Assistive effect	An improvement present while the device operates.
Retained learning	Improvement persists without stimulation after a defined interval.
Plasticity	A change in neural function or connectivity; it is not automatically beneficial or permanent.
Expert transfer	A learner policy uses information derived from expert recordings; no literal synapse copying is assumed.
State regulation	Maintaining a physiological variable or state within a target region.
Transition control	Altering when a system moves between states while preserving the intended function.
Theoretical feasibility	A causal architecture compatible with specified scientific assumptions.
Demonstrated feasibility	An experimentally observed effect within the tested conditions.

Reference and figure conventions

Bracketed numbers point to the bibliography that follows. Primary experimental papers supply the direct interface evidence. Anatomy sources and a control-theory textbook supply background; the behavioral guidance study is explicitly non-neural supporting context. No noninvasive intervention is presented as a demonstration of invasive SCCI.

All six figures are original explanatory graphics. Figures 4 and 5 contain constructed curves; none is a reproduction of participant data. Equations define assumptions or illustrate standard physical and mathematical principles. They are not validated digital twins of human spinal or autonomic function.

The paper's evidence classifications are qualitative judgments. They should not be read as numerical success probabilities, evidence of regulatory approval, or predictions of market readiness. Human pilot studies, animal mechanisms, and future hypotheses remain distinguishable throughout.

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